Journal of Neuromonitoring & Neurophysiology Vol. 4, No. 1, May 2024, 37-43
ISSN 2799-5496 https://doi.org/10.54441/jnn.2024.4.1.37

JNN

Review article

Strategies to prevent bilateral vocal fold paralysis in thyroid surgery

Dong Kun Lee, M.D., Ph.D.
Department of Otorhinolaryngology-Head and Neck Surgery,
College of Medicine, Dong-A University, Busan, Republic of Korea

Thyroid surgery carries the risk of bilateral vocal fold paralysis (VFP), which can lead to airway compromise and necessitate intervention. Despite surgical advancements, 45% of patients experience permanent paralysis, with a 30% risk of tracheotomy. Bilateral VFP can compromise the airway, presenting with noisy breathing and voice changes. Hemi-thyroidectomy carries a lower VFP risk than total thyroidectomy. IONM aids RLN preservation but hasn't consistently lowered VFP rates. Continuous IONM benefits RLN preservation but its effect on VFP rates is debated. Staged surgery post-ipsilateral nerve injury may reduce bilateral VFP risk. Preventing bilateral VFP in thyroid surgery is of paramount importance. Utilizing routine IONM and implementing staged surgery following ipsilateral nerve injury could potentially reduce the incidence of bilateral VFP. However, additional research is warranted to establish the most effective management strategies for optimizing patient outcomes in thyroid surgery.

Key words Vocal fold paralysis, Thyroidectomy, Voice, Neuromonitoring, Recurrent laryngeal nerve

Introduction

Thyroid surgery is one of the most commonly performed procedures in head and neck surgery, with the potential for excellent outcomes in the hands of experienced surgeons. However, the occurrence of bilateral VFP can significantly impact patient morbidity and mortality. Despite advancements in surgical techniques and intraoperative monitoring, bilateral VFP remains a challenging complication to manage. About 45% of Bilateral VFP patients resulting in permanent paralysis and a substantial risk of tracheotomy about 30%. Glottal area narrowing due to bilateral VFP can lead to various degrees of airway compromise, necessitating prompt intervention in 21%, overall 50% of bilateral VFP require acute airway intervention. This paper aims to explore the various aspects of bilateral VFP in thyroid surgery and discuss strategies for its prevention and management[1].

Main Text

Glottal Area Narrowing in Bilateral VFP

Glottal area narrowing refers to the reduction in the space within the glottis, the area between the vocal folds, leading to potential airway compromise. This narrowing can occur due to iatrogenic injury during surgical procedures involving the larynx, neck, thyroid

Received: 3 May, 2024. **Accepted:** 21 May, 2024

Corresponding to Dong Kun Lee
E-mail. chaos001@hanmail.net **ORCID.** https://orcid.org/0000-0002-7296-1420

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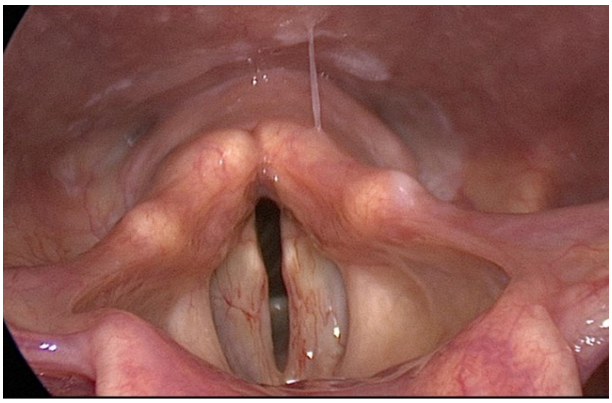


Figure 1. Laryngoscopic exam. Bilateral vocal fold paralysis with narrowed glottic area is showing.

gland, or chest. Additionally, various neurogenic conditions, such as amyotrophic lateral sclerosis and head injury, can lead to impaired vocal fold function and subsequent narrowing of the glottal area.

Patients with bilateral VFP typically experience relatively spared normal ventilation. However, they may present with various degrees of airway compromise, ranging from mild to severe. Common symptoms include noisy inspiratory breathing, primarily with minimal changes in voice quality[2](Figure 1).

The incidence of VFP varies depending on the type of thyroidectomy performed, with differences observed between hemi- and total thyroidectomy procedures. Hemithyroidectomy is associated with a significantly lower risk of temporary VFP compared to total thyroidectomy. Studies have reported temporary VFP rates of approximately 2.0% following hemi-thyroidectomy, compared to 4.2% following total thyroidectomy. The weighted risk ratio (RR) comparing the incidence of temporary VFP between hemi- and total thyroidectomy is 0.4 (95% confidence interval [CI], 0.2-0.7).

This indicates that patients undergoing hemi-thyroidectomy have a 60% lower risk of experiencing temporary VFP compared to those undergoing total thyroidectomy[3].

Does the Use of Intraoperative Nerve Monitoring (IONM) Help to Reduce the Occurrence of Bilateral VFP?

Routine Visual Identification of Recurrent Laryngeal Nerve (RLN) as Standard of Care

Routine visualization of the RLN during thyroid and parathyroid surgery is considered the standard of care. Despite meticulous anatomical dissection and identification of the nerve, RLN injury can still occur due to several factors. Variability in RLN anatomy poses challenges in consistent identification. Surgeon inexperience may contribute to inadvertent nerve injury. High-risk situations such as substernal goiter, Graves' disease, re-operative surgery, and advanced malignancy increase the complexity of RLN preservation. A significant drawback of relying solely on visualization is the inability to predict the functional integrity of a structurally intact nerve[4,5].

Value of IONM in Reducing RLN Injury

While the majority of studies have not conclusively demonstrated a reduction in the rate of VFP with the use of IONM, its role in enhancing RLN preservation is multifaceted. Despite the lack of definitive evidence showing reduced rates of nerve injury, the adoption of IONM is increasing among surgeons due to several perceived advantages. Nerve identification and neural mapping facilitate precise localization of the RLN. Functional integrity verification of the RLN aids intraoperative decision-making, enabling surgeons to modify their approach in real-time. IONM aids in elucidating the site of nerve injury and prognosticating postoperative nerve function, allowing for better patient

counseling and management[6].

The value of IONM extends beyond reducing the incidence of VFP and includes benefit for high-risk patients, support for low-volume surgeons by providing real-time feedback and guidance during surgery, addressing the variability in RLN anatomy by facilitating individualized surgical approaches, preservation of the external branch of the superior laryngeal nerve (EBSLN), particularly in type 2b, and serving as a line of defense in the event of litigation[6].

Does Continuous Vagal IONM Help to Reduce the Occurrence Rate of Bilateral VFP?

Value of Continuous IONM

Continuous vagus stimulation (CVS) has revolutionized the intraoperative surgical strategy in thyroid surgery by eliminating the lag time between nerve preparation and stimulation. This continuous monitoring empowers surgeons to be optimally aware of traction-related injury to the RLN. About 80% of loss of signal (LOS) events were caused by neural traction during IONM in a multicenter study of 115 patients[7]. Electromyographic precursor lesions, known as combined events, prompt surgeons to cease harmful maneuvers and release the nerve before irreversible damage occurs. The benefits of continuous vagal IONM are multifaceted[8].

Combined events, characterized by concordant changes in signal amplitude and latency, prompt timely intervention. In a study involving 101 patients monitored under CVS, 68% of combined events resolved after halting the causative surgical maneuver[9]. In another 102 patients enrolled study, 73% of combined events were reversible and showed positive and negative predictive values of 33% and 97%, respectively, indicating their significance in predicting VFP[10]. Among 1314 nerves at risk, 80% of combined events did not degrade to LOS upon release of the related maneuver. Discontinuation of harmful retraction led to functional recovery of all distressed nerves, particularly when LOS was yet incompleted[7].

Continuous IONM was associated with a lower frequency of adverse electromyography events (10.6 vs 20.3%, P=.001) compared to intermittent monitoring. Although there was no significant difference in intraoperative loss of signal between continuous and intermittent IONM (5.2% vs. 6.8%, P= .415), continuous monitoring showed promise in reducing adverse events. Postoperative VFP rates did not significantly differ between continuous and intermittent IONM groups (4.9 vs 7.6%, P=.168) [11]. The routine use of continuous vagal IONM improves the rate of intraoperative adverse electromyography events. However, it does not significantly impact the rates of LOS and VFP. Further research is warranted to elucidate the optimal monitoring strategies to minimize the occurrence of bilateral VFP in thyroid surgery.

Does Staged Surgery Help to Prevent Bilateral VFP after Ipsilateral Thyroidectomy?

Staged Surgery

Staged surgery, historically employed in various surgical fields to minimize complications and optimize outcomes, has gained traction in the management of

thyroid conditions. Initially utilized for exophthalmic goiter management in the 19th century, it has evolved to address both benign and malignant thyroid conditions, particularly in cases of advanced thyroid cancer. Recent research supports the use of a neural-monitored staged surgical approach to decrease locoregional complications in advanced thyroid cancer, particularly after the first surgical side[12].

Clinical Evidence for Neural Monitoring–Based Surgical Staging

Numerous surgical series have demonstrated overwhelming evidence supporting the incorporation of neural monitoring-based surgical staging into the management of thyroid conditions. Detection of LOS on the first side prompts staged bilateral thyroidectomy, leading to a significant reduction in the rate of bilateral VFP and tracheotomy. Goretzki et al. reported that the rate of bilateral VFP was 17% if LOS was undetected or disregarded despite detection, zero if the 2nd side aborted or performed as a staged procedure among 2,600 nerves at risk[13].

Value of Staged Surgery

Studies have consistently shown the value of staged surgery in preventing bilateral VFP after ipsilateral thyroidectomy. Melin et al. reported a 16% rate of bilateral VFP when surgery continued on the second side after initial LOS, compared to zero cases in staged procedures after ipsilateral LOS[14]. By their another report in 3,426 nerves at risk after ipsilateral LOS, the contralateral nerve injury rate was 28.6%[15]. Other studies by Sarkis et al., Sadowski et al., Schneider et al., Randolph et al., and Fontenot et al. corroborate these findings, demonstrating a significant reduction or absence of bilateral VFP when surgery is staged after initial LOS[7,16-19]. Additionally, Alesina et al., Thomusch et al., and Calo et al. reported an occurrence of bilateral VFP in non-monitoring arms whereas no cases in the neural monitoring cohorts[20-22].

The reasons behind the effectiveness of staged surgery in preventing bilateral VFP are multifactorial. LOS is relatively rare and indicates a complex surgical scenario, necessitating a cautious approach. Surgical complexity may extend beyond one side, increasing the risk of contralateral injury. Surgeon-related factors, including increased concern and decreased ease during contralateral surgery following an adverse event, may influence the decision to stage the procedure. Surgical complications and unexpected events can impair a surgeon's performance, emphasizing the need for meticulous planning and staged approaches[23].

IONM and Malpractice Claims of RLN Palsy

Standardized IONM has emerged as a crucial tool in preventing bilateral RLN palsies after LOS during thyroid surgery. Informed consent discussions should include the strengths and limitations of IONM, serving as a line of defense against malpractice claims. Studies have shown high patient satisfaction with staged surgery following LOS, with no significant differences between one-stage and two-stage total thyroidectomy procedures[24].

Tips for Preventing Bilateral VFP

While RLN injury during thyroidectomy is rare, it can lead to significant morbidity, emphasizing the im-

portance of meticulous surgical technique. Preservation of the superior laryngeal nerve (SLN) is also crucial due to its variable anatomy. Preoperative evaluation, including history-taking, physical examination, and imaging review, plays a pivotal role in identifying potential risk factors and planning the surgical approach[25].

General Principles

1) Meticulous hemostasis: Achieving optimal visualization through meticulous hemostasis is essential for identifying critical landmarks, including the RLN. A blood-stained surgical field can obscure anatomical structures and increase the risk of inadvertent nerve injury.
2) Utilization of advanced energy devices: Advanced energy devices are frequently used & seem to be quite effective.
3) Dissection on the thyroid capsule: Keeping dissection directly on the thyroid capsule helps minimize the risk of injury to the nerves and adjacent structures, such as the parathyroid glands. This approach reduces the likelihood of inadvertent nerve transection or thermal injury.
4) Avoidance of heat injury: Care must be taken to avoid thermal injury to the nerves, as heat generated by energy devices can cause irreversible damage to neural tissues.
5) Minimal traction on the RLN: Due to the delicate nature of the RLN, maintaining minimal traction during surgical manipulation is crucial for preventing nerve injury. Excessive traction can stretch or compress the nerve, leading to neuropraxia or axonotmesis[25].
6) Consideration of IONM: The American Academy of Otolaryngology recommends considering IONM for bilateral thyroid surgery, revision thyroid surgery, and cases of existing RLN paralysis. IONM provides real-time feedback on nerve integrity, aiding surgeons in identifying and avoiding potential nerve injury. By alerting surgeons to changes in nerve function, IONM enhances surgical precision and reduces the risk of postoperative complications, including VFP[26].

Conclusion

Efforts to prevent inadvertent ipsilateral RLN injury, despite the low rate of VFP, are imperative in thyroid surgery. Routine utilization of IONM is recommended to enhance surgical precision and minimize the risk of RLN injury. Key aspects of IONM include nerve identification, neural mapping, and verification of RLN functional integrity to aid intraoperative decision-making. When the first operation side exhibits LOS on IONM, staged surgery is recommended to reduce the risk of bilateral VFP. This approach allows for careful planning and optimization of surgical outcomes.

Acknowledgments

This study was supported by research funds from Dong-A University.

Funding

None.

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

ORCID

Dong Kun Lee : 0000-0002-7296-1420

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